

# 9. Two Stage Least Squares and Modern IV Estimators

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Jaewon (“Jay-one”) Yoo

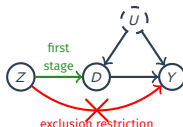
National Tsing Hua University

# Outline

1. Two-Stage Least Squares
2. Modern IV via Machine Learning
3. Estimating IV: AJR Walkthrough + Modern IV
4. Notable IV Designs in Recent Empirical Work

# Where are we? Where are we going?

- Last time:
  - **Instrumental variable** under noncompliance in randomized experiments.
    - The local ATE (or CACE):  $\tau_{\text{LATE}} = \text{ITT}_{Y,\text{co}} = \frac{\text{ITT}_Y}{\text{ITT}_D}$
    - The Wald/IV estimator:  $\widehat{\tau}_{\text{IV}} = \widehat{\text{ITT}}_Y / \widehat{\text{ITT}}_D$
    - Intent-to-treat analysis, compliance types, identification assumptions..



- Today:
  1. **Two-Stage Least Squares** (TSLS): estimands, weak IVs, multivalued  $D$ , general 2SLS.
  2. **Modern IV via Machine Learning**: PLIV (constant  $\tau$ ), DRIV (heterogeneous  $\tau(\mathbf{X})$ ), DML.
  3. **Estimating IV in practice**: AJR walkthrough + modern IV in R.
  4. **Notable IV designs**: treatment assignments, regional characteristics, peers' environments, shift-share.



Source: *Chapter 4 of Mostly Harmless Econometrics (Textbook 1) by J. Angrist & J. Pischke*

## 1/ Two-Stage Least Squares

# Two Stage Least Squares

- **Two-stage least squares** (TSLS) is the classical approach to IV which assumes two linear models:

$$Y_i = \alpha + \tau D_i + \varepsilon_i$$

$$D_i = \delta + \gamma Z_i + \eta_i$$

- Here, the treatment  $D_i$  is **endogenous** so  $\mathbb{E}[\varepsilon_i | D_i] \neq 0$ .
  - Canonical source: an omitted variable / unobserved confounder  $U$  opening a back-door path  $D \leftarrow U \rightarrow Y$ .
- But we have an **instrument**  $Z_i$  that is exogenous  $\mathbb{E}[\varepsilon_i | Z_i] = 0$ .
  - It's also exogenous for treatment uptake, so  $\mathbb{E}[\eta_i | Z_i] = 0$ .
- This implies the following CEF form for  $Y_i$  conditional on  $Z_i$ :

$$\mathbb{E}[Y_i | Z_i] = \alpha + \tau \mathbb{E}[D_i | Z_i] = \alpha + \tau \cdot (\gamma Z_i)$$

- $\alpha$  is reused loosely (technically  $\alpha + \tau\delta$  here); not the parameter of our interest.

# TSLS Estimands

- Under the model, we have the following CEF:  $\mathbb{E}[Y_i|Z_i] = \alpha + \tau \cdot (\gamma Z_i)$ 
  - $\rightsquigarrow$  A regression of  $Y_i$  on  $\gamma Z_i$  would have  $\tau$  as the slope.

- If the CEF is linear, then we have this simple relationship slopes:

$$\mathbb{E}[D_i|Z_i] = \delta + \gamma Z_i \quad \rightsquigarrow \quad \gamma = \frac{\text{cov}(D_i, Z_i)}{\mathbb{V}(Z_i)}$$

- Applying this to above CEF, we have:

$$\tau = \frac{\text{cov}(Y_i, \gamma Z_i)}{\mathbb{V}(\gamma Z_i)} = \frac{\text{cov}(Y_i, Z_i)}{\gamma \mathbb{V}(Z_i)} = \frac{\text{cov}(Y_i, Z_i)}{\text{cov}(D_i, Z_i)}$$

- TSLS estimator:
  - Estimate  $\hat{\gamma}$  from regression of treatment  $D_i$  on instrument  $Z_i$
  - Estimate  $\hat{\tau}_{\text{TSLS}}$  as the slope of a regression of  $Y_i$  on  $\hat{\gamma} Z_i$ .
    - $\hat{D}_i = \hat{\delta} + \hat{\gamma} Z_i$ ; the  $Z$ -driven  $\hat{\gamma} Z_i$  is exogenous, uncorrelated with  $\varepsilon_i$  (constant  $\hat{\delta}$  doesn't affect the slope).
  - Under this model,  $\hat{\tau}_{\text{TSLS}} \xrightarrow{P} \tau$  (but don't use SEs from second stage; see MHE section 4.6.1. 2SLS *Mistakes*)

# Binary Treatment and Instrument

- Under binary treatment/instrument, TSLS estimand is the LATE:

$$\tau = \frac{\text{cov}(Y_i, Z_i)}{\text{cov}(D_i, Z_i)} = \frac{\mathbb{E}[Y_i|Z_i = 1] - \mathbb{E}[Y_i|Z_i = 0]}{\mathbb{E}[D_i|Z_i = 1] - \mathbb{E}[D_i|Z_i = 0]} = \frac{\text{ITT}_Y}{\text{ITT}_D} = \tau_{\text{LATE}}$$

- $\text{cov}(Y, Z)/\text{cov}(D, Z) = \beta_Y/\beta_D$  ( $\mathbb{V}(Z)$  cancels, regardless of  $Z$ ). For binary  $Z$ , OLS slope  $\beta_\cdot = \mathbb{E}[\cdot|Z=1] - \mathbb{E}[\cdot|Z=0]$  (W4), giving  $\text{ITT}_Y/\text{ITT}_D$ .
- LATE defined over compliers  $\{D_i(1) > D_i(0)\}$ , single group only for binary  $D$ .
- And that the TSLS estimator is the Wald estimator:

$$\widehat{\tau}_{\text{2SLS}} = \frac{\widehat{\text{cov}}(Y_i, Z_i)}{\widehat{\text{cov}}(D_i, Z_i)} = \frac{\overline{Y}_1 - \overline{Y}_0}{\overline{D}_1 - \overline{D}_0} = \frac{\widehat{\text{ITT}}_Y}{\widehat{\text{ITT}}_D} = \widehat{\tau}_{\text{iv}}$$

↪ Constant effects model is not required for TSLS in this setting.



# TSLS with Covariates

- Adding covariates linearly:

$$Y_i = \alpha + \tau D_i + \mathbf{X}_i' \beta_y + \varepsilon_i$$

$$D_i = \delta + \gamma Z_i + \mathbf{X}_i' \beta_d + \eta_i$$

- This *uninteracted* form imposes constant  $\tau$  across  $\mathbf{X}$ . Otherwise  $\tau$  is an odd weighted function of causal effects and  $\tau \neq \tau_{\text{LATE}}$ .
- Why not Lin (W4) fully-interacted models ( $\tau + \gamma' \widetilde{\mathbf{X}}_i$  varies) in IV?
  1. Each  $D_i \widetilde{X}_{ij}$  is endogenous and requires its own instrument, e.g.,  $Z_i \widetilde{X}_{ij}$ .
  2. Strong prediction of  $D$  by  $Z$  does not imply strong prediction of  $D_i \widetilde{X}_{ij}$  by  $Z_i \widetilde{X}_{ij}$ .
  3. Many interaction terms can amplify weak-IV problems and complicate interpretation.
- $\rightsquigarrow$  **PLIV** (§2) inherits constant  $\tau$  (ML-flexible  $g(\mathbf{X})$ ,  $m(\mathbf{X})$  only). **DRIV** (§2) relaxes to  $\tau(\mathbf{X})$ .

# Weak Instruments

- IV is unstable if instrument weakly affects treatment;  $\text{cov}(D_i, Z_i) \approx 0$ .
- **Example** completely irrelevant instrument:

$$\begin{aligned} Y_i &= \tau D_i + \varepsilon_i, & \mathbb{E}[\varepsilon_i | D_i] &\neq 0 \\ D_i &= 0 \times Z_i + \eta_i, & \mathbb{E}[\varepsilon_i | Z_i] = \mathbb{E}[\eta_i | Z_i] &= 0 \end{aligned}$$

- The bias of the Wald estimator:

$$\widehat{\tau}_{\text{IV}} - \tau = \frac{\widehat{\text{cov}}(\tau D_i + \varepsilon_i, Z_i)}{\widehat{\text{cov}}(D_i, Z_i)} - \tau = \frac{\frac{1}{n} \sum_{i=1}^n \varepsilon_i Z_i}{\frac{1}{n} \sum_{i=1}^n \eta_i Z_i} \xrightarrow{d} \underbrace{\frac{\text{cov}(\varepsilon_i, \eta_i)}{\mathbb{V}[\varepsilon_i]}}_{\text{bias}} + \underbrace{W_i}_{\text{Cauchy r.v.}}$$

- Inconsistent and asymptotically heavy tails (b/c of Cauchy).
  - When  $Z \rightarrow D$  effect is small but non-zero, similar behavior.
- **Relevant IV**: noise divided by signal  $\rightarrow 0$ ; **Irrelevant IV**: noise divided by noise does not.

# What to Do About Weak Instruments?

- **Detect:** first-stage F-test on excluded instruments ( $H_0 : \gamma = 0$ ).
  - $F \geq 10 \Rightarrow$  bias is small (Stock & Yogo 2005); still the dominant convention in empirical IV.
  - $F \geq 104.7$  for correct CI coverage in the worst-case DGP (Lee et al. 2022).
    - Lee et al. show the standard 1.96 gives true 95% coverage *only* at  $F \geq 104.7$ ; below that, actual coverage drops monotonically. At the conventional  $F = 10$ , worst-case actual coverage is around **85%**.
- **Weak-IV robust inference:** Anderson-Rubin (1949) test (simplified setting, binary  $Z, D$ ).
  - **Trick:** rewrite  $H_0 : \tau = \tau_0$  as  $H_0 : \text{ITT}_Y - \tau_0 \text{ITT}_D = 0$ : **linearises the null**, no weak-IV pathology.
  - Under the null, asymptotically:

$$g(\tau_0) = \widehat{\text{ITT}}_Y - \tau_0 \widehat{\text{ITT}}_D \sim N(0, \Omega(\tau_0))$$
$$\Omega(\tau_0) = \mathbb{V}[\widehat{\text{ITT}}_Y] + \tau_0^2 \mathbb{V}[\widehat{\text{ITT}}_D] - 2\tau_0 \text{cov}(\widehat{\text{ITT}}_Y, \widehat{\text{ITT}}_D)$$

- AR statistic:  $g(\tau_0)^2 / \Omega(\tau_0) \sim \chi_1^2$  regardless of first-stage strength; CI by analytical inversion.

```
1 > library(ivmodel)
2 > m <- ivmodel(Y = y, D = D, Z = Z)
3 > m$AR      # AR test (F-stat), p-value, df, and 95% CI by inversion
```

- Increasingly reported alongside the F-stat; handles weak-IV uncertainty correctly when first stage is borderline.

# Multi-Valued Treatments

- Generalization of these ideas:
  - Multi-valued treatment:  $D_i \in \{0, 1, \dots, K - 1\}$
  - Binary instrument:  $Z_i \in \{0, 1\}$
- Assumptions:
  - Randomization:  $[\{Y_i(z, d), \forall z, d\}, D_i(1), D_i(0)] \perp\!\!\!\perp Z_i$
  - Monotonicity:  $D_i(1) \geq D_i(0)$  (instrument only increases treatment)
  - Exclusion restriction:  $Y_i(1, d) = Y_i(0, d)$  for all  $d = 0, 1, \dots, K - 1$
- Can't identify the proportion of all compliance types here.
- Example:  $K = 3 \rightsquigarrow 9$  principal strata
  - Affected:  $(D_i(0), D_i(1)) \in \{(0, 1), (0, 2), (1, 2)\}$
  - Unaffected:  $(D_i(0), D_i(1)) \in \{(0, 0), (1, 1), (2, 2)\}$
  - Negatively affected:  $(D_i(0), D_i(1)) \in \{(1, 0), (2, 0), (2, 1)\}$
  - Last ruled out by monotonicity.

# TSLS with Multivalued Treatments: ACR

- Under the assumptions on the previous slide, Angrist–Imbens 1995, JASA, Theorem 1:

$$\widehat{\tau}_{2SLS} \xrightarrow{p} \sum_{j=1}^{K-1} \omega_j \cdot \mathbb{E} \left[ Y_i(j) - Y_i(j-1) \mid D_i(1) \geq j > D_i(0) \right]$$
$$\omega_j = \frac{\mathbb{P}[D_i(1) \geq j > D_i(0)]}{\mathbb{E}[D_i(1) - D_i(0)]}, \quad \sum_{j=1}^{K-1} \omega_j = 1.$$

- Intuition: a weighted average of effects per dose for each affected type.
  - Weights are proportional to size of the strata and how big the effect of the instrument is for that strata.
  - If instrument can only increase by 1 dose, then simplifies to weighted average of principal strata effects.
- This object is the **Average Causal Response (ACR)**, the multi-valued analogue of LATE (named in MHE Th 4.5.3). Binary case ( $K = 2$ ) collapses to standard LATE.

## 2/ Modern IV via Machine Learning

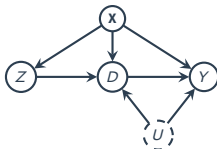
Cf. Chernozhukov et al. (2018), *Double/Debiased Machine Learning*, Econometrics Journal

# From Classical TSLS-with- $\mathbf{X}$ to Modern IV

- Empirical IV studies routinely need to condition on a large set of confounders, and increasingly involve **unstructured / high-dim  $\mathbf{X}$**  (text, image embeddings, many interactions). Social-science theory does not pin down which features and which functional form to use.
- Classical TSLS-with- $\mathbf{X}$  enters  $\mathbf{X}$  linearly with a constant  $\tau$ . Two limits given the above:
  - Linear/parametric  $\mathbf{X}$  leaves nuisance specification fragile.
  - Adding  $\mathbf{D} \cdot \mathbf{X}$  interactions for HTE drops you into the Winston Lin fully-interacted IV trap (recall §1: weak instruments, sign flips, multiple LATEs).
- Two-step ladder out:
  - **PLIV** (Robinson 1988; Chernozhukov et al. 2018 adds DML): keep constant  $\tau$ ; let ML handle  $\mathbf{X}$  flexibly. *Same  $\tau$ , cleaner nuisance.*
  - **DRIV** (Syrkanis et al. 2019, NeurIPS): drop constant  $\tau$ ;  $\tau(\mathbf{X})$  ML-learned. *HTE built into the spec.*
- Both run on the **Double/Debiased ML** engine: cross-fitting + Neyman-orthogonal moment + ML for the three nuisance CEFs  $\mathbb{E}[Y|\mathbf{X}]$ ,  $\mathbb{E}[D|\mathbf{X}]$ ,  $\mathbb{E}[Z|\mathbf{X}]$ .

# ML Nuisance Functions and High-Dim Confounding

- Why ML for IV:  $\mathbf{X}$  is high-dim (text, image embeddings, many interactions) and theory specifies neither features nor functional form.
- DAG with covariates  $\mathbf{X}$  and unobserved confounder  $U$ :



- **Three nuisance functions** to estimate:

$$\ell_0(\mathbf{X}) = \mathbb{E}[Y \mid \mathbf{X}], \quad r_0(\mathbf{X}) = \mathbb{E}[D \mid \mathbf{X}], \quad m_0(\mathbf{X}) = \mathbb{E}[Z \mid \mathbf{X}]$$

- When  $\mathbf{X} \not\rightarrow Z$  (i.e.,  $Z \perp \mathbf{X}$ ),  $m_0$  collapses to the constant  $\mathbb{E}[Z]$  and ML has nothing to do; it becomes a real function of  $\mathbf{X}$  only when  $\mathbf{X}$  predicts  $Z$ .
- **Why naive plug-in fails:** ML regularization injects bias into  $\hat{\eta}$  that contaminates  $\hat{\tau}$ . DML fixes this with two ideas:
  - **Neyman orthogonality:** moment is locally insensitive to nuisance errors at the truth.
  - **Cross-fitting:** fit  $\hat{\eta}$  on out-of-fold data, evaluate on held-out fold.



# PLIV via DML: Algorithm and Conditions

- **Model:**  $Y = \tau D + g(\mathbf{X}) + \varepsilon$ ,  $\mathbb{E}[\varepsilon \mid \mathbf{X}, Z] = 0$ .
- **Algorithm** (non-linear, high-dim Frisch-Waugh-Lovell):  $K$ -fold cross-fitting.
  1. Partition into folds  $\{I_k\}_{k=1}^K$ . For each  $k$ , fit  $\hat{\ell}_{[k]}, \hat{r}_{[k]}, \hat{m}_{[k]}$  on data *outside*  $I_k$ .
  2. For  $i \in I_k$ :  $\tilde{Y}_i = Y_i - \hat{\ell}_{[k]}(\mathbf{x}_i)$ ,  $\tilde{D}_i = D_i - \hat{r}_{[k]}(\mathbf{x}_i)$ ,  $\tilde{Z}_i = Z_i - \hat{m}_{[k]}(\mathbf{x}_i)$ .
  3. Solve  $\mathbb{E}_n[\tilde{Z}_i(\tilde{Y}_i - \tau \tilde{D}_i)] = 0$ : **standard IV** on residuals.
- **Validity conditions** (CHS 2018, Thm 13.1.1): strong IV after partialling out, plus ML estimates converging at  $n^{1/4}$  rate.
- Software: DoubleML (R). Applied PLIV demo on AJR in §3.

# DRIV: Letting $\tau$ Depend on $\mathbf{X}$

- **One-symbol difference from PLIV:**

PLIV:  $Y = \theta \cdot D + g(\mathbf{X}) + \varepsilon$  (constant  $\theta$ )

DRIV:  $Y = \theta(\mathbf{X}) \cdot D + g(\mathbf{X}) + \varepsilon$  ( $\theta(\mathbf{X})$  ML-learned)

HTE built into the spec.

- **Why TSLS-with- $\mathbf{X}$  cannot:** estimating  $\theta(\mathbf{X})$  would require instrumenting  $D \cdot \mathbf{X}$  with  $Z \cdot \mathbf{X}$ , the Winston Lin fully-interacted IV trap (recall §1: weak-IV, sign flips, multi-LATE).
- **How DRIV does it** (Syrgkanis et al. 2019, NeurIPS): same residualization as PLIV, but the orthogonal IV moment is solved *pointwise at each  $\mathbf{X}$* :

$$\theta(\mathbf{X}) = \frac{\mathbb{E}[\tilde{Z} \tilde{Y} \mid \mathbf{X}]}{\mathbb{E}[\tilde{Z} \tilde{D} \mid \mathbf{X}]}, \quad \text{learned by ML across } \mathbf{X}.$$

PLIV averages this conditional Wald ratio over  $\mathbf{X}$  to get one  $\theta$ ; DRIV keeps the conditioning.

Doubly-robust orthogonality ensures first-order ML errors in  $\hat{\ell}, \hat{r}, \hat{m}$  do not propagate to  $\hat{\theta}(\mathbf{X})$ .

- **R cousin:** `grf::instrumental_forest` (Causal Forest IV) – different moment (forest-weighted local 2SLS), same target  $\tau(\mathbf{X})$ , same conclusions in practice.

```
1 # DRIV in R via reticulate (Python EconML under the hood)
2 library(reticulate); econml_dr <- import("econml.iv.dr"); sk <- import("sklearn.ensemble")
3 est <- econml_dr$DRIV(
4   model_y_xw = sk$RandomForestRegressor(), # E[Y|X]
5   model_t_xw = sk$RandomForestClassifier(), # E[D|X]
6   model_z_xw = sk$RandomForestClassifier(), # E[Z|X]
7   cv = 5L)
8 est$fit(Y, T = D, Z = Z, X = X); est$effect(X_new) # tau-hat(X)
```

# Choosing the ML Learner: How Sensitive Is DML?

- Nuisance estimation needs a learner choice (Lasso, RF, XGBoost, neural net) and tuning. Different choices can yield very different DML point estimates on the same data.
- Ahrens, Chernozhukov, Hansen, Kozbur, Schaffer, Wiemann (2025) re-examine Dube, Jacobs, Naidu, Suri (2020) (AER, online labor monopsony,  $n \approx 258k$ ): **18 candidate learners**  $\Rightarrow$  **point estimates from  $-0.39$  to  $+0.15$ .**
- **Two remedies they propose:**
  - **CVC test** (Lei 2020, *Biometrika*):  $H_0$  : “this learner has the lowest predictive risk among candidates.” Keep learners with  $p > 0.1 \sim 0.2$ ; report DML using that subset.
  - **Stacking / model averaging**: data-driven weighted average over the full set of candidate learners (*super learner*; Wolpert 1992; van der Laan, Polley, Hubbard 2007). Single point estimate, no manual learner pick.
- **Practical**: ddm1 R package implements both. Report cross-fitted  $R^2$  for each nuisance regression as a sanity check before using its DML estimate.

# Open Question: X Selection Itself

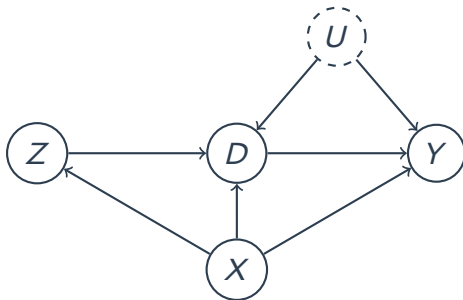
- DML / DRIV assumes the analyst hands the algorithm a “right” set of  $\mathbf{X}$  to condition on. **Which  $\mathbf{X}$ ?** is itself a causal-graph question.
- **Canonical pitfall:** conditioning on a *collider* opens a spurious  $D$ - $Y$  path that does not exist marginally. Classical FWL/OLS practice catches this via DAG reasoning before any regression (recall ECI W6).
- **ML cannot recognize colliders:** a tree or Lasso fitting  $\mathbf{X} \rightarrow Y$  uses any predictive signal, including signal operating through a collider. Variable importance is *not* causal validity. See [Hünermund, Louw & Caspi \(2023, JCI\)](#).
- **Active research direction:** combining ML’s prediction strength with causal-graph reasoning so that the analyst (or the algorithm) does not silently introduce collider bias when assembling  $\mathbf{X}$ . *Work in progress.*

## 3/ Estimating IV: AJR Walkthrough + Modern IV

# From Theory to Practice: Classical TSLS Setup

- Modern IV (§2) shines when  $\mathbf{X}$  is high-dim or text-scale. AJR sits in the **classical corner**: a single binary IV, modest  $\mathbf{X}$ ; classical TSLS suffices.
- Classical-corner ingredients:
  1. observational data with *unobserved* confounders  $U$ ;
  2. a single instrumental variable  $Z$  with a qualitatively defensible exclusion restriction;
  3. low-dim observed covariates  $\mathbf{X}$ ; (linearly conditioned).
- DAG (recap from §2): see next slide.

# Classical TSLS Setup: DAG (recap from §2)



- $Z \rightarrow D$ : first stage.  $D \rightarrow Y$ : causal effect of interest.  $U$ : unobserved confounder.  $X$ : observed covariates.
- Exclusion restriction:  $Z$  affects  $Y$  only through  $D$  (no direct  $Z \rightarrow Y$  arrow).

# Property Rights & Economic Development



Created using DALL-E 3 motivated by Ratna Sagar Shrestha



Recognize the person on the right?

- Q: Do property rights (i.e., institutions) promote economic development?
  - Famous paper on this: **Acemoglu, Johnson, and Robinson (2001) AER**
  - Relationship between strength of property rights in a country and GDP.



# The AJR Data

| Name     | Description                                                    |
|----------|----------------------------------------------------------------|
| shortnam | three-letter country code                                      |
| africa   | indicator for if the country is in Africa                      |
| asia     | indicator for if country is in Asia                            |
| logem4   | log mortality rates faced by European settlers (IV)            |
| avexpr   | strength of property rights (protection against expropriation) |
| logpgp95 | log GDP per capita                                             |

```
1 > ajr <- read_csv("https://bit.ly/3RUJDWK"); ajr
2
3 # A tibble: 163 × 15
4   shortnam africa lat_abst malfal94 avexpr logpgp95 logem4 asia yellow baseco leb95
5   <chr>      <dbl>   <dbl>   <dbl> <dbl>   <dbl>   <dbl> <dbl> <dbl> <dbl> <dbl>
6 1 AFG          0 0.367 0.00372 NA      NA      4.54 1 0 NA NA
7 2 AGO          1 0.137 0.950 5.36 7.77 5.63 0 1 1 46.5
8 3 ARE          0 0.267 0.0123 7.18 9.80 NA 1 0 NA NA
9 4 ARG          0 0.378 0 6.39 9.13 4.23 0 0 1 72.9
10 5 ARM          0 0.444 0 NA 7.68 NA 1 0 NA NA
11 6 AUS          0 0.300 0 9.32 9.90 2.15 0 1 1 78.2
12 7 AUT          0 0.524 0 9.73 9.97 NA 0 0 NA NA
13 8 AZE          0 0.448 0 NA 7.31 NA 1 0 NA NA
14 9 BDI          1 0.0367 0.950 NA 6.57 5.63 0 1 NA NA
15 10 BEL          0 0.561 0 9.68 9.99 NA 0 0 NA NA
16 # i 153 more rows
17 # i 4 more variables: imr95 <dbl>, meantemp <dbl>, lt100km <dbl>, latabs <dbl>
18 # i Use `print(n = ...)` to see more rows
```

# In R: Example Code Using ajr Data

```
1 # Center (i.e., demeaning) the variables
2 > ajr <- ajr |>
3   mutate(
4     D_cnt = avexpr - mean(avexpr, na.rm = TRUE),
5     Y_cnt = logpgp95 - mean(logpgp95, na.rm = TRUE),
6     Z_cnt = logem4 - mean(logem4, na.rm = TRUE)
7   ) |>
8   na.omit()
9
10 # Compute the ITTs on D and on Y:
11 > ITT_D <- cov(ajr$Z_cnt, ajr$D_cnt)
12 > ITT_Y <- cov(ajr$Z_cnt, ajr$Y_cnt)
13 > wald_estimate <- ITT_Y / ITT_D; round(wald_estimate, digits = 4)
14 [1] 0.9242
15
16 # Same as reg Y~Z / reg D~Z:
17 > ITT_Y <- coef(lm(Y_cnt ~ Z_cnt, data = ajr))[[2]]
18 > ITT_D <- coef(lm(D_cnt ~ Z_cnt, data = ajr))[[2]]
19 > wald_estimate <- ITT_Y / ITT_D; round(wald_estimate, digits = 4)
20 [1] 0.9242
```

- The Wald estimate from manual calculation: 0.9242

# In R: Example Code Using ajr Data

```
1 # Compare with ivreg
2 > ivreg_result <- AER::ivreg(Y_cnt ~ D_cnt | Z_cnt, data = ajr); summary(ivreg_result)
3
4 Call:
5 AER::ivreg(formula = Y_cnt ~ D_cnt | Z_cnt, data = ajr)
6
7 Residuals:
8      Min       1Q   Median       3Q      Max
9 -2.40175 -0.54950  0.01792  0.68944  1.67361
10
11 Coefficients:
12             Estimate Std. Error t value Pr(>|t|)
13 (Intercept) 3.031e-16  1.231e-01   0.000      1
14 D_cnt       9.242e-01  1.547e-01   5.974 1.59e-07 ***
15 ---
16 Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
17
18 Residual standard error: 0.9458 on 57 degrees of freedom
19 Multiple R-Squared: 0.1107,      Adjusted R-squared: 0.09506
20 Wald test: 35.68 on 1 and 57 DF, p-value: 1.589e-07
21
22 > round(coef(ivreg_result), digits = 4)
23 (Intercept)      D_cnt
24      0.0000      0.9242
```

- The 2sls estimate from AER::ivreg(): 0.9242
  - Caveat: **compute robust SE separately!** (don't use SE from 2nd stage)

# In R: Example Code Using ajr Data

```
1 # Estimate robust SE with estimatr::iv_robust()
2 > iv_rob <- estimatr::iv_robust(Y_cnt ~ D_cnt | Z_cnt, data = ajr, se_type = "HC2"); iv_rob
3           Estimate Std. Error      t value      Pr(>|t|)      CI Lower CI Upper DF
4 (Intercept) 4.148820e-16  0.1234908 3.359619e-15 1.000000e+00 -0.2472860 0.247286 57
5 D_cnt       9.242173e-01  0.1791511 5.158871e+00 3.262823e-06  0.5654735 1.282961 57
```

- Robust SE with HC2 using `estimatr::iv_robust()`: 0.1791

```
1 > iv_rob_cov <- estimatr::iv_robust(Y_cnt ~ D_cnt + lat_abst + asia + africa |
2                                   Z_cnt + lat_abst + asia + africa, data = ajr,
3                                   se_type = "HC2")
4 > var_labels <- c(
5   "D_cnt" = "Avg. Expropriation Risk (D_cnt)", "lat_abst" = "Abs. Value of Latitude",
6   "asia" = "Asian country", "africa" = "African country", "Z_cnt" = "log Mortality Rate (D_cnt)"
7 )
```

- When adjusting for covariates in TSLS, include them both in the 1st and 2nd stage.

# IV Regression Table with model summary

```
1 > modelsummary::modelsummary(  
2   models = list(ivreg_result, iv_robust, iv_rob_cov),  
3   coef_map = var_labels, gof_map = c("nobs", "r.squared", "adj.r.squared"), stars = T,  
4   notes = "Note: See appendix for other model statistics.", output = "modelsummary_tab.tex")
```

|                                 | (1)                 | (2)                 | (3)                |
|---------------------------------|---------------------|---------------------|--------------------|
| Avg. Expropriation Risk (D_cnt) | 0.924***<br>(0.155) | 0.924***<br>(0.179) | 1.015**<br>(0.351) |
| Abs. Value of Latitude          |                     |                     | -1.596<br>(1.529)  |
| Asian country                   |                     |                     | -1.048*<br>(0.425) |
| African country                 |                     |                     | -0.390<br>(0.342)  |
| Num.Obs.                        | 59                  | 59                  | 59                 |
| R2                              | 0.111               | 0.111               | 0.045              |
| R2 Adj.                         | 0.095               | 0.095               | -0.026             |

+ p < 0.1, \* p < 0.05, \*\* p < 0.01, \*\*\* p < 0.001

Note: See appendix for other model statistics.

# Modern IV in R: DoubleML for PLIV

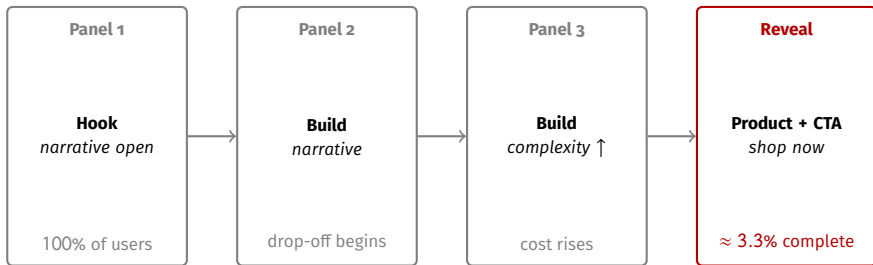
- Same partialling-out as Modern IV section, in ~10 R lines (cross-fitting built in):

```
1 > library(DoubleML); library(mlr3); library(mlr3learners)
2 > # Build DoubleMLData (y = outcome, d = treatment, z = instrument, x = covariates)
3 > data <- DoubleMLData$new(df, y_col = "Y", d_cols = "D",
4 +                           z_cols = "Z", x_cols = X_names)
5 > # ML learners for the nuisance functions
6 > ml_l <- lrn("regr.ranger") # E[Y | X]
7 > ml_m <- lrn("regr.ranger") # E[Z | X]
8 > ml_r <- lrn("regr.ranger") # E[D | X]
9 > # PLIV with 5-fold cross-fitting (Neyman orthogonality + cross-fitting = DML)
10 > pliv <- DoubleMLPLIV$new(data, ml_l, ml_m, ml_r, n_folds = 5)
11 > pliv$fit(); pliv$summary()
```

- Heterogeneous  $\tau(\mathbf{X})$  in R: `grf::instrumental_forest` (Athey–Tibshirani–Wager 2019). See §2 for learner-selection and  $\mathbf{X}$ -selection caveats.

# What Does a BLI Ad Look Like?

- Blind Lead-In (BLI) ads are multi-stage interactive sliders. The product is revealed *only* on the last panel; users must advance through each panel themselves.



- In passive media (TV, pre-roll), exposure  $\Rightarrow$  reveal automatically. In BLI, only  $\approx 3.3\%$  of exposed users reach the reveal.
- $\Rightarrow$  The behavioral object of interest is  $D$  (completion), not  $Z$  (exposure). Standard ITT-on- $Z$  averages over 96.7% never-takers.

# Modern IV Demo: A Digital Advertising Experiment

- Online book retailer A/B-tests two ad formats ( $n \approx 1.4\text{M}$  user-impressions, 76 ad creatives).
  - $Z$ : random assignment to interactive (Blind Lead-Ins, BLI) vs. static banner.
  - $D$ : BLI *completion* (treated-side completion rate  $\approx 3.32\%$ ; **one-sided noncompliance**:  $D = 0$  for all  $Z = 0$ ).
  - $Y$ : 30-day purchase conversion (binary).
- Why this is a useful DML-IV testbed:
  - High-dim creative features (face presence, slide count, character count, color, area ratios).
  - Rich user covariates (spending, visits, age, gender, B2B, genre fit).
  - Plausible  $\tau(\mathbf{X})$  heterogeneity along creative-content  $\times$  user-engagement axes.
- **Plan**: (i) classical pooled 2SLS LATE (fixest); (ii) DRIV via reticulate (econml.LinearIntentToTreatDRIV) for  $\widehat{\theta}(\mathbf{X})$ ; (iii) Causal Forest IV (grf::instrumental\_forest) for nonparametric  $\widehat{\tau}(\mathbf{X})$ . *Two distinct HTE-IV estimators on the same data; do they agree?*



# Pooled 2SLS LATE: Classical Baseline

- 2SLS LATE on 30-day conversion (date FE, user covariates, cluster-robust SE at the user level):

```
1 > library(fixest)
2 > late <- feols(lngt_conv ~ ctrl_vars | date | tot ~ treatment,
3 +             data = d, cluster = ~ memno)
4 > summary(late)
5 TSLS estimation, Dep. Var.: lngt_conv
6 Observations: 1,408,121 / Fixed-effects: date (80)
7 Standard-errors: Clustered (memno)
8             Estimate Std. Error t value Pr(>|t|)
9 fit_tot 0.002326   0.001210   1.922   0.0548 .
10 ... (control coefficients omitted)
11 First-stage F (Kleibergen-Paap): ~1.5e4 # very strong instrument
```

- **LATE:** BLI completion raises 30-day conversion by  $\approx 0.23$  **percentage points** among compliers ( $\text{fit\_tot} = 0.00233$ ,  $p \approx 0.055$ ).
- **Weak-IV diagnostic:** first-stage  $F = 10,007$  (Kleibergen-Paap, cluster-robust). Compliance is small (3.32%) but big-tech ad RCTs are typically large- $N$  enough that the first stage is overwhelming; Stock-Yogo  $F \geq 10$  **vastly** cleared.

# DRIV via Reticulate (Syrkanis et al. 2019)

- DRIV: heterogeneous LATE  $\theta(X) = \alpha + \beta'X$  via cross-fitted nuisance + doubly-robust orthogonal score. Implementation:  
`econml.LinearIntentToTreatDRIV` (Python only). Call from R via `reticulate`:

```
1 library(reticulate)
2 econml_iv <- import("econml.iv.dr"); sk <- import("sklearn.ensemble")
3 est <- econml_iv$LinearIntentToTreatDRIV(
4   model_y_xw = sk$RandomForestRegressor(n_estimators = 100L),
5   model_t_xwz = sk$RandomForestClassifier(n_estimators = 100L),
6   flexible_model_effect = "auto", cv = 5L, cov_clip = 0.005)
7 est$fit(Y = Y, T = D, Z = Z, X = X_paper, W = W_paper)
8 est$ate__inference()$summary_frame()
```

- ATE across `cov_clip` sensitivity (key trimming knob for low-compliance cells):

| cov_clip | ATE     | s.e.    | 95% CI             | Note                |
|----------|---------|---------|--------------------|---------------------|
| 0.001    | 0.00197 | 0.00122 | [-0.0004, +0.0044] | most permissive     |
| 0.005    | 0.00247 | 0.00107 | [+0.0004, +0.0046] | <b>book default</b> |
| 0.010    | 0.00209 | 0.00078 | [+0.0006, +0.0036] |                     |
| 0.050    | 0.00056 | 0.00019 | [+0.0002, +0.0009] | over-trimmed        |

- Linear projection  $\beta'X$  flags which features drive  $\hat{\theta}(X)$  heterogeneity. With face-stage decomposition (`first_has_face`, `middle_has_face`, `late_has_face`): `middle_has_face` is the only significant face term  $\Rightarrow$  middle-panel face matters causally. Side-by-side with Causal Forest IV in the next slide.

# From $\hat{\tau}$ to $\hat{\tau}(\mathbf{X})$ : Causal Forest IV in R

- R-native HTE-IV (distinct from DRIV; §2): `grf::instrumental_forest` (Athey–Tibshirani–Wager 2019);  $\mathbf{X}$  = 4 user + 4 creative features.

```
1 > library(grf)
2 > ivf <- instrumental_forest(
3 +   X = X_matrix, Y = Y, W = D, Z = Z,
4 +   num.trees = 4000, min.node.size = 100, honesty = TRUE)
5 > average_treatment_effect(ivf)      # forest-weighted ATE on BLI sample
6 > predict(ivf, X_new)$predictions    # tau-hat(X) at any covariate point
7 > best_linear_projection(ivf, A = X_matrix) # which features drive HTE; valid SE
8 > test_calibration(ivf)              # is HTE detectable above noise?
```

- **Causal Forest ATE**:  $\approx 0.17$  pp on average (forest-weighted; conservative vs. pooled LATE because low-compliance cells are down-weighted).
- **BLP** marginal HTE drivers: `face_early` ( $p \approx 0.07$ ), `pref_matched` ( $p \approx 0.10$ )  $\Rightarrow$  creative-content  $\times$  user-engagement axes.
- Same target  $\hat{\tau}(\mathbf{X})$  as DRIV (§2), built from forest-weighted local 2SLS rather than the orthogonal-score moment.

# What HTE-IV Reveals on the BLI Data

- **Pooled-LATE ladder converges:** 2SLS 0.00229, DML PLIV (Lasso) 0.00217, DML IIVM (RF) 0.00164 ( $p = 0.049$ ), DRIV (econml, cov\_clip=0.005) 0.00247. All in the  $[+322\%, +484\%]$  lift band  $\Rightarrow$  the BLI-completion effect on conversion is robust to ML, score, and functional-form choice.
- **DRIV and Causal Forest IV both flag middle-panel face presence:**
  - middle\_has\_face (positive): Causal Forest BLP  $\hat{\beta} = +0.0144$  ( $p = 0.008$ ); DRIV linear  $\hat{\beta} = +0.0118$  ( $p = 0.014$ ). *Both significant.*
  - late\_has\_face (negative): Causal Forest  $\hat{\beta} = -0.0103$  ( $p = 0.086$ , marginal); DRIV  $\hat{\beta} = -0.005$  ( $p = 0.33$ , not significant). The forest picks up the late-stage penalty more sharply.
  - Engaged users (high recent visit-days) absorb  $\approx 0.39$  pp LATE; disengaged  $\approx 0.05$  pp  $\Rightarrow$  near-8 $\times$  heterogeneity gap pooled 2SLS hides.
- **Mapping back to the paper:** H4 (face presence raises completion) confirmed at the *conversion-LATE* level. Stage asymmetry (middle dominant, late negative) is now *causally identified* on the downstream outcome, not just a descriptive completion-stage association.

# Lift Translation: from $\hat{\tau}$ to Industry Magnitude

- Gordon, Zettelmeyer, Bhargava & Chapsky 2019, *Marketing Science* eq. (8): lift = incremental conversion among treated, expressed as a percentage of the treated group's *counterfactual* (untreated) conversion.

$$\tau_{\ell} = \frac{\Delta \text{Conversion}_{\text{treated}}}{\mathbb{E}[Y^{obs} \mid Z = 1, W^{obs} = 1] - \tau} = \frac{\tau}{\mathbb{E}[Y^{obs} \mid Z = 1, W^{obs} = 1] - \tau}$$

- Under random assignment, the denominator is identified by the control-group conversion. NNE =  $1/\text{ITT}_Y = 1/(\text{compliance rate} \times \hat{\tau}_{\text{LATE}})$ : ad impressions per incremental purchase.
- BLI lift table (control conversion 0.051% as estimator of treated-counterfactual baseline):

| Method                               | $\hat{\theta}$ | Compliers conv. | Lift (%) | NNE    |
|--------------------------------------|----------------|-----------------|----------|--------|
| 2SLS pooled (paper)                  | 0.00229        | 0.280%          | +449%    | 13,200 |
| DML PLIV (Lasso)                     | 0.00217        | 0.268%          | +426%    | 13,900 |
| DML IIVM (RF)                        | 0.00164        | 0.215%          | +322%    | 18,400 |
| <b>DRIV (econml, cov_clip 0.005)</b> | 0.00247        | 0.298%          | +484%    | 12,300 |

- Industry benchmark: typical online display-ad NNE is 50,000–200,000. BLI's 12,300–18,400 is order-of-magnitude better.
- Back-of-envelope:** 100M annual impressions  $\times$  3.32% completion  $\times \hat{\tau}_{\text{LATE}} (\approx 0.0023) \approx 7,640$  **extra purchases/year** causally attributable to BLI completion ( $\approx 114\text{M}$  KRW at the retailer's average book price).

## 4/ Notable IV Designs in Recent Empirical Work

**When you have a great idea for an IV but the first stage turns out to be not significant.**

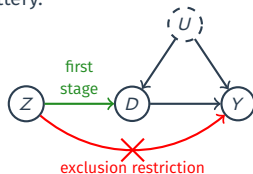


Source: [Causal Inference for the Brave and True](#) By Matheus Facure Alves

# Treatment Assignments as IVs

## Bloom et al. (QJE 2015)<sup>1</sup>

- Context: 9-month RCT in Ctrip's Shanghai call-centre; 249 volunteers entered the lottery.



Instrument Z: even-birthday lottery eligibility.

Treatment D: actually working from home  $\geq 4$  days/week.

Outcome Y: weekly productivity index (calls, minutes logged-in, calls/minute).

- Identification issue: Endogenous uptake; one-sided non-compliance (some winners revert, losers cannot WFH).
- IV logic:
  1. Randomization: Balance tests on 18 baseline characteristics show no joint differences;  $p=0.466$ .
  2. Relevance: 80-90% of winners comply (1st-stage  $F \approx 23$ )
  3. Exclusion: tasks, pay, IT identical – only location changes.
  4. Monotonicity: winners may or may not take WFH, losers cannot  $\rightsquigarrow$   
 $D_i(1) \geq D_i(0)$  trivially (one-sided non-compliance).

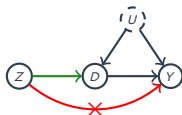
<sup>1</sup> Bloom, N., Liang, J., Roberts, J. & Ying, Z. "Does Working from Home Work?" QJE 2015. 130(1): 165–218.



# Regional Characteristics as IVs

## Narang and Shankar (*Marketing Sci.* 2019)<sup>2</sup>

- Context: Observational panel on 32 mil. customers of a U.S. video-game / electronics retailer, Jan 2013 – Dec 2015; firm's mobile-shopping app launched July 2014.



Instrument Z: number of cell towers in shopper's ZIP.

Treatment D: adoption of the retailer's mobile app.

Outcome Y: Monthly on/offline purchase & return amounts.

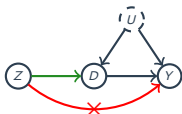
- Identification issue: Self-selection. Tech-savvy or high-value shoppers more likely to install the app.
- IV logic:
  1. Randomization: 90% of cell-towers built before 2010 and shows only weak correlations with 2010–15 population growth and store count, included in  $\mathbf{X}_i$ .
  2. Relevance: Each cell tower raises the log-odds of adoption by 0.0022 ( $z=3.7$ ).
  3. Exclusion: controlled for ZIP income, store distance, and competitor presence to block direct demand channels.
  4. Monotonicity: Better signal only increases the prob. of downloading the app (no defiers).

<sup>2</sup> Narang & Shankar. "Mobile App Introduction and On- /Off-line Purchases and Product Returns." *Marketing Sci.* 2019. 38(5): 756 – 772.

# Peers' Environments as IVs

## Aral and Nicolaides (*Nature Comm.* 2017)<sup>3</sup>

- Context: Global fitness-tracking network, 1.1 Mil. runners, 3.4 Mil. ties, 350 Mil. km logged over 5 years.



Instrument Z: daily weather (rain/extreme temperature) in friend's city.

Treatment D: friend's running distance that day.

Outcome Y: user/ego's running distance (same or next day).

- Identification issue: Homophily & shared shocks. Similar friends exercise together or face the same local weather.
- IV logic:
  1. Randomization: Use only friend pairs in different cities whose weather paths are uncorrelated; ego's own weather + date FE included.
  2. Relevance: first-stage  $F = 216-430$  well above Stock-Yogo cutoff.
  3. Exclusion: Weather in friend's city is uncorrelated with ego's weather by construction.
  4. Monotonicity: bad weather never makes the friend run more.

<sup>3</sup> Sinan A. & Nicolaides C. "Exercise contagion in a global social network." *Nature Communications*. 2017. 8: 14753.

## Onto the presentations & discussions!

*Contact Information:*

[jaewon.yoo@iss.nthu.edu.tw](mailto:jaewon.yoo@iss.nthu.edu.tw)

<https://j1yoo.github.io/>



# Appendix

# BLI Demo: Method-Comparison Table

| Method                              | $\hat{\theta}$ | s.e.    | 95% CI              | Notes                                    |
|-------------------------------------|----------------|---------|---------------------|------------------------------------------|
| 2SLS pooled (paper)                 | 0.00229        | 0.00121 | [-0.00004, 0.00469] | fixest, date FE, cluster=memno           |
| DML PLIV (Lasso)                    | 0.00217        | 0.00116 | [-0.00010, 0.00444] | Book Ch 13.1 score, $n_{\text{rep}} = 5$ |
| DML PLIV (RF, m=featureless)        | 0.00137        | 0.00190 | [-0.00235, 0.00509] | Book Ch 13.1                             |
| DML IIVM (Lasso)                    | 0.00217        | 0.00116 | [-0.00009, 0.00444] | Book Ch 13.2                             |
| <b>DML IIVM (RF)</b>                | 0.00164        | 0.00083 | [+0.00001, 0.00328] | <b>p = 0.049</b>                         |
| DRIV linear [cov_clip 0.001]        | 0.00197        | 0.00122 | [-0.00041, 0.00435] | econml.LinearIntentToTreatDRIV           |
| <b>DRIV linear [cov_clip 0.005]</b> | 0.00247        | 0.00107 | [+0.00037, 0.00458] | <b>book default</b>                      |
| DRIV linear [cov_clip 0.010]        | 0.00209        | 0.00078 | [+0.00057, 0.00362] |                                          |

- All point estimates in [0.00164, 0.00247], lift band [+322%, +484%] over 0.051% control conversion.
- Working paper identification + inference still under revision; values shown are illustrative for the lecture demo.

# BLI Demo: Subgroup CATE (engaged vs disengaged users)

| Subgroup                           | <i>n</i> | CATE    |
|------------------------------------|----------|---------|
| Engaged (high recent visit-days)   | 686,903  | 0.00222 |
| Disengaged (low recent visit-days) | 721,218  | 0.00114 |
| Age $\geq 30$                      | 932,236  | 0.00146 |
| Age $< 30$                         | 475,885  | 0.00207 |
| face_early present                 | 586,313  | 0.00235 |
| face_early absent                  | 821,808  | 0.00118 |

- Engaged  $\times$  face-early users absorb *nearly all* the BLI completion lift; disengaged users with no early face: near-zero CATE.
- Source: `grf::instrumental_forest` subgroup CATE; trimmed cells with compliance  $< 0.5\%$ .

# General 2SLS

- Linear model for each  $i$ :

$$Y_i = \mathbf{X}_i' \boldsymbol{\beta} + \varepsilon_i$$

- $\mathbf{X}_i$  is  $k \times 1$  and now includes  $D_i$  and any pretreatment covariates.
- Parts of  $\mathbf{X}_i$  are endogenous so that  $\mathbb{E}[\varepsilon_i | \mathbf{X}_i] \neq 0$
- Instruments  $\mathbf{Z}_i$  that is  $\ell \times 1$  vector such that  $\mathbb{E}[\varepsilon_i | \mathbf{Z}_i] = 0$ .
  - $\mathbf{Z}_i$  might include exogenous/pretreatment variables from  $\mathbf{X}_i$  as well.
  - Rank condition:  $\mathbb{E}[\mathbf{Z}_i \mathbf{Z}_i']$  and  $\mathbb{E}[\mathbf{X}_i \mathbf{Z}_i']$  have full rank.
- Identification:
  - $k = \ell$ : just-identified.
  - $k < \ell$ : over-identified (can test the exclusion restriction, kinda)
  - $k > \ell$ : unidentified (fails rank condition)

# Nasty Matrix Algebra

- Projection matrix projects values of  $\mathbf{X}_i$  onto  $\mathbf{Z}_i$ :

$$\mathbf{\Pi} = (\mathbb{E}[\mathbf{Z}_i \mathbf{Z}_i'])^{-1} \mathbb{E}[\mathbf{Z}_i \mathbf{X}_i'] \quad (\text{projection matrix, i.e., } \mathbf{\Pi} / \mathbf{\Pi})$$

$$\tilde{\mathbf{X}}_i = \mathbf{\Pi}' \mathbf{Z}_i \quad (\text{projected values})$$

- To derive the 2SLS estimator, take the fitted values,  $\mathbf{\Pi}' \mathbf{Z}_i$  and multiply both sides of the outcome equation by them:

$$Y_i = \mathbf{X}_i' \beta + \varepsilon_i$$

$$\mathbf{\Pi}' \mathbf{Z}_i Y_i = \mathbf{\Pi}' \mathbf{Z}_i \mathbf{X}_i' \beta + \mathbf{\Pi}' \mathbf{Z}_i \varepsilon_i$$

$$\mathbb{E}[\mathbf{\Pi}' \mathbf{Z}_i Y_i] = \mathbb{E}[\mathbf{\Pi}' \mathbf{Z}_i \mathbf{X}_i'] \beta + \mathbb{E}[\mathbf{\Pi}' \mathbf{Z}_i \varepsilon_i]$$

$$\mathbb{E}[\mathbf{\Pi}' \mathbf{Z}_i Y_i] = \mathbb{E}[\mathbf{\Pi}' \mathbf{Z}_i \mathbf{X}_i'] \beta + \mathbf{\Pi}' \mathbb{E}[\mathbf{Z}_i \varepsilon_i]$$

$$\mathbb{E}[\mathbf{\Pi}' \mathbf{Z}_i Y_i] = \mathbb{E}[\mathbf{\Pi}' \mathbf{Z}_i \mathbf{X}_i'] \beta$$

$$\mathbb{E}[\tilde{\mathbf{X}}_i Y_i] = \mathbb{E}[\tilde{\mathbf{X}}_i \mathbf{X}_i'] \beta$$

$$\beta = (\mathbb{E}[\tilde{\mathbf{X}}_i \mathbf{X}_i'])^{-1} \mathbb{E}[\tilde{\mathbf{X}}_i Y_i]$$



# How to Estimate the Parameters

- Collect  $\mathbf{x}_i$  into an  $n \times k$  matrix  $\mathbb{X} = (\mathbf{x}'_1, \dots, \mathbf{x}'_n)$
- Collect  $\mathbf{z}_i$  into an  $n \times \ell$  matrix  $\mathbb{Z} = (\mathbf{z}'_1, \dots, \mathbf{z}'_n)$
- In-sample projection matrix produces fitted values:

$$\widehat{\mathbb{X}} = \mathbb{Z}(\mathbb{Z}'\mathbb{Z})^{-1}\mathbb{Z}'\mathbb{X}$$

- Fitted values of the regression of  $\mathbb{X}$  on  $\mathbb{Z}$ .
- Matrix party trick:  $\mathbb{X}'\mathbb{Z}/n = (1/n) \sum_i^n \mathbf{x}_i \mathbf{z}'_i \xrightarrow{P} \mathbb{E}[\mathbf{x}_i \mathbf{z}'_i]$ .
- Take the population formula for the parameters:

$$\beta = (\mathbb{E}[\tilde{\mathbf{x}}_i \mathbf{x}'_i])^{-1} \mathbb{E}[\tilde{\mathbf{x}}_i y_i]$$

- And plug in the sample values (the  $n$  cancels out):

$$\widehat{\beta}_{2SLS} = (\widehat{\mathbb{X}}'\mathbb{X})^{-1}\widehat{\mathbb{X}}'\mathbf{y} \xrightarrow{P} \beta$$

- This is how R/Stata estimate the 2SLS parameters.

# Asymptotic Variance for 2SLS

- We can write the centered, normalized TSLS estimator as:

$$\sqrt{n}(\widehat{\beta}_{2SLS} - \beta) = \underbrace{\left(n^{-1} \sum_i \widehat{\mathbf{x}}_i \widehat{\mathbf{x}}_i'\right)^{-1}}_{\xrightarrow{p} (\mathbb{E}[\widehat{\mathbf{x}}_i \widehat{\mathbf{x}}_i'])^{-1}} \underbrace{\left(n^{-1/2} \sum_i \widehat{\mathbf{x}}_i \varepsilon_i\right)}_{\xrightarrow{d} N(0, \mathbb{E}[\widehat{\mathbf{x}}_i' \varepsilon_i' \varepsilon_i \widehat{\mathbf{x}}_i])}$$

- Thus,  $\sqrt{n}(\widehat{\beta}_{2SLS} - \beta)$  has asymptotic variance:

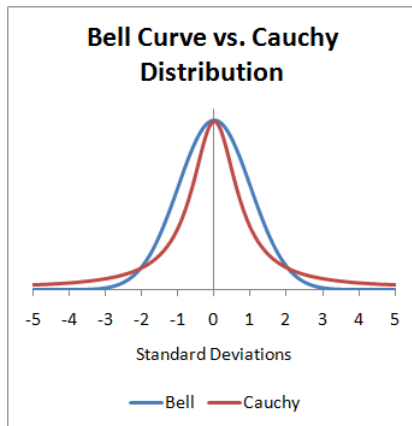
$$(\mathbb{E}[\widehat{\mathbf{x}}_i \widehat{\mathbf{x}}_i'])^{-1} \mathbb{E}[\widehat{\mathbf{x}}_i' \varepsilon_i' \varepsilon_i \widehat{\mathbf{x}}_i] (\mathbb{E}[\widehat{\mathbf{x}}_i \widehat{\mathbf{x}}_i'])^{-1}$$

- Robust 2SLS variance estimator** with residuals  $\widehat{u}_i = Y_i - \mathbf{x}_i' \widehat{\beta}$ :

$$\widehat{\text{Var}}(\widehat{\beta}_{2SLS}) = (\widehat{\mathbb{X}}' \widehat{\mathbb{X}})^{-1} \left( \sum_i \widehat{u}_i^2 \widehat{\mathbf{x}}_i \widehat{\mathbf{x}}_i' \right) (\widehat{\mathbb{X}}' \widehat{\mathbb{X}})^{-1}$$

- HC2, clustering, and autocorrelation versions exist.

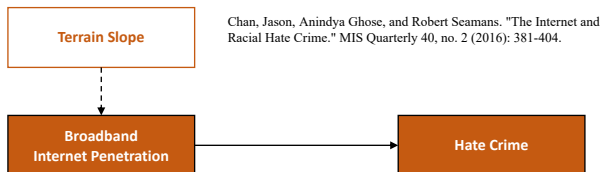
# Cauchy vs. Normal Distribution



Source: <https://stats.stackexchange.com/questions/36027/why-does-the-cauchy-distribution-have-no-mean>

# Regional Characteristics as IVs

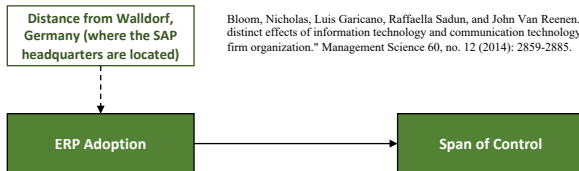
- Chan, Ghose, Seamans, MISQ, 2016:



- Observational study using *Hate Crime Statistics* from FBI.
  - RQ: Does the spread of Internet increase racial hate crime?
- Issue? Confounding
- IV? terrain slope/steepness (i.e., how many hills in a given region?)

# Geographical Proximity-Based IVs

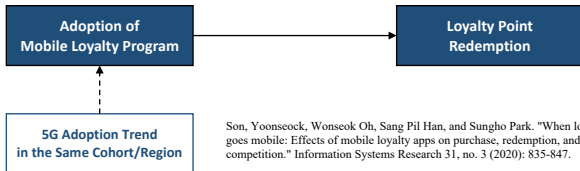
- RQ: Does the adoption of an ERP system impact plant manager autonomy (i.e., span of control)?
  - Span of control: the number of employees managed by supervisors or managers in an organization (high SoC = centralized).



- Observational study: the CEP management and organization survey and the Harte-Hanks ICT panel (Bloom, Garicano, Sadun, Van Reenen, *MgmtSci*, 2014).
- IV: Distance from the ERP market leader (i.e., SAP) w/ 25%+ market share  $\rightsquigarrow$  likely more established connections with SAP (German firms vs. French, England firms).

# Macro/Cohort Trends as IVs

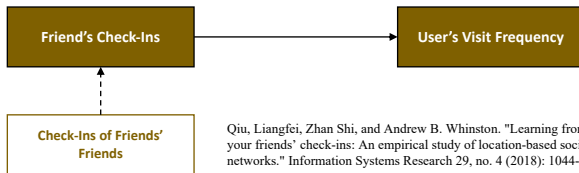
- RQ: How does loyalty app adoption affect customer point redemption behavior?



- Observational study: data on loyalty app adoption status, loyalty point redemption patterns, and purchase behaviors in multivendor loyalty program (MVLN) context (Son, Oh, Han, Park, ISR, 2020).
- IV: 5G adoption rate in the same cohort (e.g., age group).

# Peers' Environments as IVs

- Qiu, Shi, Whinston, ISR, 2018:

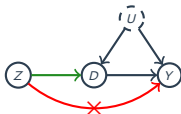


- Observational study using data on restaurant check-in information and the users' social network ties from a major SNS in China.
  - RQ: Is there observational learning/herding effect for restaurant discovery?
- Issue? Homophily.
- IV: Check-in activities of friend's friends.

# Bartik (Shift-Share) IVs

## Barron et al. (*Marketing Sci.* 2020)<sup>4</sup>

- Context: Panel of 43,000 ZIP codes (100 largest U.S. metro areas), monthly 2011-2016. Airbnb listings scraped; Zillow rent & price indices matched at ZIP-month.



Instrument Z: Google-Trends “Airbnb” (global shock)  $\times$  2010 ZIP “touristiness” (# food/lodging firms)

Treatment D:  $\ln(1 + \text{Airbnb listings})$  in ZIP-month.

Outcome Y: Zillow rent index, house price index, price-to-rent.

- Identification issue: Hot ZIPs attract both Airbnb supply and rising housing costs  $\rightsquigarrow$  upward bias.
- IV logic:
  - Conditional exogeneity: 2010 touristiness fixed before Airbnb’s national growth; Google-Trends shock is global, not driven by ZIP-specific factors. **(Hardest assumption for Bartik IVs; defended in detail next slide.)**
  - Relevance: first-stage  $F = 650\text{--}820$  well above Stock-Yogo cutoff.
  - Exclusion: global search shocks unrelated to local housing; touristiness fixed in 2010. City $\times$ month FE + placebo ZIPs (no listings) show no direct price effect.
  - Monotonicity: more global Airbnb awareness cannot lower listings in touristy ZIPs.

<sup>4</sup>Barron, K., Kung, E., & Proserpio, D. “The Effect of Home-Sharing on House Prices and Rents” *Marketing Sci.* 2020. 40(2): 283–304.



# Bartik (Shift-Share) IVs

## Barron et al. (*Marketing Sci.* 2020)

- IV logic 1 (continued): defending **conditional exogeneity**, the hardest assumption for Bartik IVs:
  - Parallel pre-trends: prior to 2012, rents and prices evolve identically across touristiness quartiles (Fig. 5)  $\rightsquigarrow$  no pre-existing divergence.
  - Touristiness  $\times$  time test: adding a ZIP-specific trend term ( $h_{i,2010} \times t$ ) yields an insignificant coefficient; IV effect unchanged.
  - Rich time-varying controls: results robust after adding ZIP income, population, hotel rooms, TripAdvisor reviews, and airport arrivals  $\rightsquigarrow$  IV not picking up gentrification or tourism shocks.